Standard Operating Procedure

Calibration Verification Procedure

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1. Purpose

This procedure defines the calibration verification process for NEXUS immunoassay kits during manufacturing QC release testing. It specifies the method for measuring calibrator signals (S1, S2), acceptance criteria, and error code definitions.

2. Scope

Applicable to QC calibration verification testing of all NEXUS products at the Marcy-l'Etoile facility. This SOP covers the use of NEXUS calibrators S1 (high concentration) and S2 (low concentration / negative control) on NEXUS analyzers.

3. Equipment Required

Equipment ID	Description	Use
NEXUS-001 / NEXUS-002	NEXUS Analyzer	Primary measurement instrument
EQ-TEMP-001	Calibrated thermometer	Ambient temperature verification
N/A	Product-specific calibrators (S1, S2)	Reference standards
N/A	Product-specific SPR cones + strips	Test consumables

4. Procedure

4.1 Preparation

4.1.1 Verify that the NEXUS analyzer has been powered on for at least 30 minutes and ambient temperature is 20-25 C.
4.1.2 Verify the analyzer calibration status is current (last PM within 3 months).
4.1.3 Retrieve calibrators S1 and S2 from 2-8 C storage. Allow to equilibrate to room temperature for 15 minutes.
4.1.4 Record lot numbers for: SPR cones, reagent strips, calibrators S1 and S2 on the QC record form.

4.2 Calibrator Loading

4.2.1 Reconstitute calibrators as per product-specific IFU.
4.2.2 Pipette calibrator S1 into designated wells (positions 1-2).
4.2.3 Pipette calibrator S2 into designated wells (positions 3-4).
4.2.4 Load SPR cones and reagent strips into the NEXUS analyzer.
4.2.5 Start the calibration verification programme.

4.3 Signal Measurement and Recording

4.3.1 After programme completion (~35 minutes), record the S1 and S2 signal values in RFV (Relative Fluorescence Value).
4.3.2 Calculate the mean of duplicate measurements for each calibrator.
4.3.3 Record any error codes generated by the analyzer.

5. Acceptance Criteria

Parameter	Specification	Measurement	Error Code if OOS
S1 (High Calibrator)	450 - 550 RFV	Mean of duplicates	B18 if outside range
S2 (Low Calibrator)	80 - 120 RFV	Mean of duplicates	B18 if outside range
S1/S2 Ratio	>= 3.5	Calculated	B19 if below threshold
Duplicate CV	<= 5%	Calculated	B20 if exceeded

WARNING: S1 signal must be within 450 - 550 RFV. Values exceeding the upper limit (> 550 RFV) indicate possible SPR coating issue or calibrator degradation. Values below lower limit (< 450 RFV) indicate possible reagent strip issue or instrument malfunction.

6. Error Code Definitions

Code	Name	Description	Action
B18	Calibration Invalid	S1 or S2 signal outside specification range. Calibration curve cannot be established.	Retest with fresh calibrators. If retest fails, quarantine lot and d
B19	S1/S2 Ratio Low	Insufficient discrimination between high and low calibrators. May indicate reagent degradation.	Retest with new reagent strip. If persists, investigate strip lot.
B20	Duplicate CV Exceeded	Poor reproducibility between duplicate measurements. May indicate pipetting error or instrument issue.	Retest with careful pipetting technique. If persists, check inst

7. Troubleshooting Guide

Symptom	Possible Cause	Investigation Steps
S1 too high (> 550 RFV)	SPR coating issue	Check SPR lot COA for antibody density (should be 2.0-2.5 mg/mL). Compare with pre
S1 too high (> 550 RFV)	Calibrator degradation	Verify calibrator storage conditions (2-8 C). Check expiry date. Retest with new calibrat
S1 too low (< 450 RFV)	Reagent strip issue	Check reagent strip lot. Verify storage conditions. Retest with strip from different lot if a
S1 too low (< 450 RFV)	Instrument malfunction	Check instrument PM status. Run system diagnostics. Verify optical sensor with referen
S2 out of range	Instrument or reagent issue	Retest on different NEXUS analyzer. If S2 fails on both instruments, investigate reagen

8. Out-of-Specification (OOS) Handling

8.1 On first OOS result: retest with fresh calibrators (same lot).
8.2 If retest confirms OOS: quarantine the production lot.

8.3 Open NCR per SOP-QA-010 within 24 hours of confirmed OOS.
8.4 Notify Production Manager and Quality Manager.
8.5 Do NOT release any lot with confirmed B18 error code.
8.6 Investigate per 5M methodology (Milieu, Matiere, Methode, Main d'oeuvre, Machine).

9. Revision History

Rev	Date	Author	Changes
1	2020-01-15	J. Martin	Initial release
2	2021-03-15	J. Martin	Added duplicate CV check and error code B20
3	2022-12-01	J. Martin	Updated troubleshooting guide; added SPR coating check per INV-2022-0045 findings

10. Approval

Role	Name	Signature	Date
Author	J. Martin		
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